

Algebra: Partial Fractions & Division



Polynomial long division

- 1 Divide $x^3 + 2x^2 - 5x + 3$ by $x - 2$, stating the quotient and remainder.
 - 2 Divide $2x^3 - 3x^2 + 4x - 1$ by $x + 1$, stating the quotient and remainder.
 - 3 Divide $x^4 - 1$ by $x^2 + 1$, stating the quotient and remainder.
 - 4 Divide $3x^3 + 5x^2 - 2x + 7$ by $x + 2$, stating the quotient and remainder.
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Partial fractions: distinct linear factors

- 5 Express $\frac{7x - 4}{(x - 1)(x + 3)}$ in partial fractions.
 - 6 Express $\frac{2x + 5}{x(x + 5)}$ in partial fractions.
 - 7 Express $\frac{9 - x}{(x + 2)(x - 4)}$ in partial fractions.
 - 8 Express $\frac{4x - 1}{(x - 2)(x + 1)}$ in partial fractions.
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Partial fractions: repeated linear factors

- 9 Express $\frac{3x + 2}{(x + 2)^2}$ in partial fractions of the form $\frac{A}{x + 2} + \frac{B}{(x + 2)^2}$.
 - 10 Express $\frac{x - 4}{(x - 1)^2}$ in partial fractions of the form $\frac{A}{x - 1} + \frac{B}{(x - 1)^2}$.
 - 11 Express $\frac{2x + 3}{(x - 3)^2}$ in partial fractions of the form $\frac{A}{x - 3} + \frac{B}{(x - 3)^2}$.
 - 12 Express $\frac{5x^2 - 4x - 1}{x(x - 1)^2}$ in partial fractions of the form $\frac{A}{x} + \frac{B}{x - 1} + \frac{C}{(x - 1)^2}$.
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Partial fractions: irreducible quadratic factors

- 13 Express $\frac{4x^2 - x + 3}{(x - 2)(x^2 + 1)}$ in partial fractions of the form $\frac{A}{x - 2} + \frac{Bx + C}{x^2 + 1}$.
- 14 Express $\frac{2x^2 + 3}{(x + 1)(x^2 + 2)}$ in partial fractions of the form $\frac{A}{x + 1} + \frac{Bx + C}{x^2 + 2}$.
- 15 Express $\frac{x^2 + 2}{(x + 2)(x^2 + 4)}$ in partial fractions of the form $\frac{A}{x + 2} + \frac{Bx + C}{x^2 + 4}$.

Improper fractions: combining division with partial fractions

- 16 Express $\frac{x^3 + 1}{x^2 - 1}$ as a polynomial plus a proper fraction, simplifying fully.
- 17 Express $\frac{x^2 + 3x + 1}{x + 1}$ as a polynomial plus a proper fraction, using division.
- 18 Express $\frac{2x^3 - x^2 + 3}{x^2 - 1}$ as a polynomial plus a proper fraction, simplifying fully.
- 19 Express $\frac{x^2 + 5x + 2}{x + 3}$ as a polynomial plus a proper fraction, using division.
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The cover-up method

- 20 Use the cover-up method to express $\frac{5x^2 + x - 2}{(x - 1)(x + 2)(x - 4)}$ in partial fractions.
- 21 Use the cover-up method to express $\frac{7x - 3}{x(x - 1)(x + 3)}$ in partial fractions.
- 22 Use the cover-up method to express $\frac{3x + 5}{(x + 1)(x - 2)}$ in partial fractions.
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Using partial fractions for series expansion

- 23 Express $\frac{3}{(1 - x)(1 + 2x)}$ in partial fractions, then write the first three terms of its series expansion in ascending powers of x .
- 24 Express $\frac{2}{(1 + x)(1 - 2x)}$ in partial fractions, then find the coefficient of x^2 in its series expansion in ascending powers of x .
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Verifying decompositions by equating coefficients

- 25 Given $\frac{2x + 1}{(x + 1)(x + 3)} \equiv \frac{A}{x + 1} + \frac{B}{x + 3}$, find A and B by equating coefficients, and verify using $x = 0$.
- 26 Given $\frac{x + 7}{x^2 - x - 6} \equiv \frac{A}{x - 3} + \frac{B}{x + 2}$, find A and B by equating coefficients, and verify using $x = 1$.
- 27 Given $\frac{x - 1}{(x + 2)(x - 3)} \equiv \frac{A}{x + 2} + \frac{B}{x - 3}$, find A and B by equating coefficients, and verify using $x = 0$.
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A well-designed word problem: an electrical transfer function

- 28 This question has three parts. An engineer analyzes a circuit with transfer function $H(s) = \frac{5s + 7}{(s + 1)(s + 2)}$.
 (a) Express $H(s)$ in partial fractions. (b) Find $H(0)$ directly from the original expression, and verify it matches the value from your decomposition. (c) A modified circuit has transfer function $\frac{5s^2 + 12s + 9}{(s + 1)(s + 2)}$.
 Using polynomial division, express this in the form (constant) + (partial fractions).